EDWIN SETSOAFIA KWADZO

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ABOUT ME

I am a very hardworking, disciplined, and self-motivated first-class graduate of Mechatronics Engineering with a passionate interest in pursuing a career in research in the field of embodied intelligence and Human-Robot Interaction (HRI) in order to contribute to addressing the limited ability of robots to proactively understand human intentions, learn through physical interaction, and adapt effectively within dynamic real-world environments. I believe with Robotics and AI, we can build an eco-friendly world where machines can co-exist with humans. I see Robots as tools that extends our capabilities as humans with biological limitations. In this era of artificial brain development, these pages showcase academic/industrial/personal projects, aiding me in acquiring the necessary skills and contribution to the future I have always envisioned.

EDUCATION

Koforidua Technical University *(completed)* - Koforidua, Ghana — August 2021 – May 2025
Address - 3P8P+F5F, Koforidua-Nsutam Rd, Koforidua, Ghana (https://ktu.edu.gh)
Bachelor of Technology in Mechatronics Engineering - Final Grade: 4.36/5.0 (Top 1% of over 3000 graduated students)
Relevant Courses (Grades) - C/C++ Programming 1 & 2 (A+), Embedded Systems 1 &2 (A+), Calculus 1 & 2 (A+), Control systems 1 & 2 (A), Electronic Product Development (A), Digital Signal and Image Processing (A+), Research Methods (B+), Mechatronic actuators (A), Robotics (A+), Computer Network Communication (A+), System Design Engineering (A+)

WORK EXPERIENCE

Kasapreko PLC, Graduate Intern (Maintenance Engineer Mechanical/Automation) — December 2025 – Present
- Assist in the preventive and corrective maintenance of Krones production line equipment, including Blow Moulders, Fillers, Palletizers, RoboBox systems, Variopac packers, Helix shrink wrappers, and conveyor systems to ensure high equipment availability.
- Participate in scheduled preventive maintenance activities, including inspection, lubrication, alignment, component replacement, calibration, and functional testing.
- Support troubleshooting and fault diagnosis of mechanical, electrical, pneumatic, and control system failures to minimize production downtime.
- Assist with equipment overhauls during planned shutdowns by dismantling, inspecting, repairing, reassembling, and testing machine components.
- Perform routine inspections of sensors, actuators, motors, gearboxes, bearings, pneumatic systems, and conveyor assemblies to ensure reliable operation.
- Monitor machine performance and identify potential equipment issues before they result in failures or production interruptions.
- Collaborate with maintenance engineers and technicians to resolve production line breakdowns using systematic fault-finding techniques and technical documentation.
- Record maintenance activities, equipment faults, spare parts usage, and corrective actions in accordance with plant maintenance procedures.
- Adhere to safety, quality, and Good Manufacturing Practice (GMP) standards while carrying out maintenance activities in a high-speed beverage production environment.
- Contribute to improving equipment reliability and operational efficiency by supporting continuous improvement and maintenance initiatives.

Fortress E&P Ghana, Student Intern (Robotics) — October 2024 – December 2024
- Contributed to the development of a Visitor Log System backend (using python) for use in Fortress Ghana.
- Documented my research and development work on the Visitor Log System in a form of report using Microsoft Word.
- Assisted with the testing of ROV (Remotely Operated Vehicle) equipment and machinery for offshore operations.
- Assisted engineers with the calibration of DC Motors for ROV (Remotely Operated Vehicle) for research and development.

ACADEMIC PROJECTS

Koforidua Technical University, mbed API — June 2025 – September 2025
- Led a team of four (4) to develop Application Programming Interface (API) for a custom STM32 microcontroller which runs on an ARM Cortex M0+ Processor. The API uses wrapper classes similar to Arm Mbed OS abstracting lower-level more complex hardware interface for students taking a course in Embedded Systems at Koforidua Technical University. The API handles the underlying implementation details, allowing students to focus on logic rather than register manipulation. The API now renamed to DarkLight have been refactored and ported to other microcontrollers to enhance teaching and learning of embedded systems in Koforidua Technical University.

Koforidua Technical University, Bachelor Thesis I & II January 2025 – October 2025

- Supervised individual team contribution as the team leader for five (5) other Mechatronics students.
- Performed literature review on Glass Crushing Technology for recycling.
- Identified gaps in the underlying research area.
- Proposed a system architecture and control algorithm based on research findings.
- Simulated a Low Pass Butterworth filter in python using Google Colab which was later ported to C/C++ code and deployed on an Arm Cortex M0 microcontroller.
- Simulated PID control algorithm in MATLAB for deployment on an Arm Cortex M0 microcontroller.
- Led firmware architecture design and integration of event-triggered/time-triggered real-time control tasks.
- Developed and tested PID control algorithms with real-time current feedback for crushing motor torque regulation.
- Designed and implemented safety interlocks including limit switch-based gate control with fault detection and emergency stops.
- Developed Human-Machine Interface (HMI) with LCD display, LEDs, and push buttons enabling intuitive user control.
- Conducted unit tests on key firmware modules to ensure robustness and fault tolerance.
- Documented research work as a project report using LaTeX, Mendeley and Microsoft Word.

Koforidua Technical University, Robotics Module June 2024 – October 2024

- Supervised individual team contribution as the team leader for three (3) other Mechatronics students.
- Implemented Forward & Inverse Kinematics (FK & IK) for a 4-DoF Robotic Arm on an Arduino Uno R3(C/C++).
- Programmed USB Camera to identify and recognize objects in the Robots workspace using OpenCv-Python.
- Performed test procedures to ensure the Robotics Arm worked as expected.

Koforidua Technical University, Product Development Module June 2023 – October 2023

- Supervised individual team contribution as the team leader for six (6) other Mechatronics students.
- Developed C/C++ libraries that were used to program a Digital Clock Product.
- Programmed an Electronic Digital Clock Product in C/C++ on an ARM Cortex $M0^+$ microcontroller.
- Performed test procedures to ensure optimal functionality and reliability of the software.
- Assisted Electronics team with soldering of the individually addressable segments of the APA107 LED Strip.

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

Aspire Leaders Program (Aspire Institute & Harvard Faculty) - *Team Leadership, Analytical Skills and Project Management Skills.*
Skill Lync - *Introduction to Finite Element Analysis*
Currently Developing:

- PCB Fabrication and Testing
- ROS 2(Robot Operating System) Fundamentals
- Advanced Control Systems for Robotics
- Reinforcement Learning for Robotics

SKILLS

Competencies/Soft skills:	*I always approach every problem before me with analytical reasoning because of my creative and innovative nature. My time management skills make me an organized individual helping me to meet project deadlines, milestones, and always willing to take on new challenges. Above all am a team player and detail-oriented individual.*
Electrical/Electronics:	*Circuit Design, PCB Soldering, PCB Design, Ki-CAD EDA, MATLAB Simulink, Simscape Multibody, Control System Design and Modelling, Sensor Integration.*
Mechanical:	*Autodesk Inventor Professional, CAD Modelling & Design, Basic FEA, Addictive Manufacturing (3D printing).*
Embedded Systems:	*ARM Cortex $M0^+$ Microcontrollers, Segger Embedded Studio, STM32CubeIDE, Visual Studio Code*
Programming:	*C, C++, Python, MATLAB*
Computer Vision:	*Open CV*
Research Tools:	*Microsoft word with Mendeley, LaTeX (Academic Level)*

EXTRA CURRICULAR

Innovate Koforidua Challenge, *Assistant Coach (Koforidua. Sec. Tech.)*, Eastern Tech. Hub October 2025

- Assisted in mentoring Koforidua Secondary Technical students for a technical innovation challenge.
- Supervised the design and construction of a Smart Water Dispenser.

EchoNote, Speech-to-text desktop application, *Personal Project* April 2024

- Designed the user interface (GUI) using flet a python GUI framework.
- Coded python methods to dynamically handle button-click events.
- Coded methods to export the resulting text to PDF format with an export button.

5V Linear Voltage Power Supply, PCB *Design, Personal Project* November 2023

- Performed schematic capture using KiCAD EDA.
- Assigned appropriate foot-print to various components in the design.
- Performed electrical rule check and rectified errors.
- Performed efficient power routing for the 2-layer PCB.

Custom Arduino Uno R3, PCB *Design, Personal Project* October 2023

- Performed schematic capture using KiCAD EDA.
- Assigned appropriate foot-print to various components in the design.
- Performed electrical rule check and rectified errors.
- Performed power routing, signal routing and differential pair routing on the 2-layer PCB.

Robotics Club, *CAD Team Lead*, Koforidua Technical University June 2022 – October 2022

- Designed, modelled (using Autodesk Inventor) and 3D printed battery casing and a compartment unit for sensors and circuitry that was assembled in a smart bin system.

HONOURS & AWARDS

First Class Honours Award in Mechatronics Engineering, KTU December 2025
MTN Bright Scholarship Award for Academic Excellence, MTN Ghana January 2024
Aspire Leaders Program Award for completion of leadership program, Aspire Institute October 2025

PROFESSIONAL MEMBERSHIP/AFFILIATIONS

ASPIRE LEADERS PROGRAM, *Alumnus*, Aspire Institute & Havard Faculty October 2025 – Present

- Developed leadership, critical thinking, problem-solving, and strategic decision-making skills through interactive coursework and collaborative projects.
- Collaborated with a diverse international cohort, strengthening cross-cultural communication, teamwork, and global networking skills.
- Enhanced professional competencies in leadership, ethics, career development, and effective communication through workshops led by academic and industry experts.
- Engaged in discussions on innovation, entrepreneurship, and creating sustainable impact within local and global communities.

MTN BRIGHT SCHOLARSHIP, *Alumnus*, MTN Ghana October 2025 – Present

- Selected as an MTN Bright Scholar based on outstanding academic performance and leadership potential.
- Participated in leadership development, mentorship, and career readiness programs designed to foster future industry and community leaders.
- Collaborated with fellow scholars in professional networking and personal development activities.

TUESAG-KTU, *Member*, Koforidua Technical University January 2022 – August 2025

Technical University Engineering Students Association of Ghana – Koforidua Technical University

Welcome to my E-portfolio to learn more about my projects, experiences, certifications and awards.